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$$\begin{aligned}\int \frac{4x+7}{2x^2-3x-2} dx &= \int \frac{4x+7}{(2x+1)(x-2)} dx \\ \frac{4x+7}{(2x+1)(x-2)} &= \frac{A}{2x+1} + \frac{B}{x-2} \\ A = \frac{4x+7}{x-2} \Big|_{x=-\frac{1}{2}} &= -2 \\ B = \frac{4x+7}{2x+1} \Big|_{x=2} &= 3 \\ \frac{4x+7}{2x^2-3x-2} &= \frac{-2}{2x+1} + \frac{3}{x-2} \\ \Rightarrow -2 \int \frac{d(2x+1)}{2x+1} + 3 \int \frac{d(x-2)}{x-2} \\ &= -\ln|2x+1| + 3\ln|x-2| + c\end{aligned}$$

References